Appendix to Ripple: Measuring the Cost of Over-Helping in Robot Fleets

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This appendix holds the material the paper refers to but has no room for: the system prompt every model call used, the one text for each event kind, one complete live call and one veto prompt with their replies, and the per-run counts for the 40 model runs. The study code is the delivery crate of the Ripple repository, and the prompt and call texts here are the files beside this document, as the crate wrote them.

I. The frozen system prompt

Every model call in the study used this system prompt. It was written once, before any scored run, and never changed. The mission file holds it.

```
You are the fleet manager for a regional blood delivery
     service. A classical dispatcher assigns each request to
     the idle vehicle at the site's own hub that has the
     stock and the range and arrives soonest; it does not
     read events. Fixed-wing aircraft drop by parachute and
     cannot land; multirotors can also land on hospital
     rooftops, which take only a landing, and have a short
     range. Each hub holds stock of each blood type. Sites
     are named by hub number and letter: 3A to 3F are hub 3's
     clinics, 3H1 and 3H2 its hospitals. You never assign
     single requests: you give orders that change what the
     dispatcher may do, where vehicles and stock are, and
     what comes first. Keep trauma and urgent deliveries on
     time, keep flights out of closed areas, and do not send
     vehicles beyond their endurance. Reply with JSON only,
     no other text.
```

II. Event texts

Each event kind has one text, written once and filled in from the event's fields. The model reads events as these sentences under *New events* and *Events still in force* (Section III). The placeholders are the hub or hubs, the site, the stock item, the area, the vehicle class, and the start and end minutes.

Kind	Text
Surge	A mass-casualty incident near {hubs}: the hospitals there are sending trauma requests beyond what those hubs can fly alone, starting at minute {from}.
Shortage	{item} has run out at {hubs} as of minute {from}. Requests for {item} at their sites can only be served from hubs that still hold it.
Closure	Airspace is closed in {area} from minute {from} until minute {to}; no flights through it.
Outage	{hubs} cannot launch from minute {from} until minute {to}; its sites must be covered by other hubs.
Weather	Weather at {hubs}: no {class} launches from minute {from} until minute {to}.
Priority	Hospital {site} takes priority over routine deliveries from minute {from} until {to}.
Rooftop closed	The rooftop at hospital {site} is closed from minute {from} until minute {to}; only drops at its ground pad are allowed.
Corridor	A medevac helicopter corridor is active in {area} from minute {from} until minute {to}; no vehicle may cross it below 150 m.

III. One complete model call

The prompt below is one call to Opus as a live manager with the exceptions view and feedback on a 16-hub day, at minute 35, and the reply that came back. The prompt is what the delivery crate wrote; nothing was removed. The order format at the end is part of every live prompt. The reply closed the rooftop that had just been reported and moved two fixed-wing aircraft from hub 5 to hub 4, the one hub with waiting requests and no idle vehicle, with a return time.

```
Time: minute 35 of the day; requests stop coming at minute
     240.

New events:
- The rooftop at hospital 16H2 is closed from minute 35 until
     minute 155; only drops at its ground pad are allowed.

Events still in force:
- The rooftop at hospital 3H2 is closed from minute 10 until
     minute 130; only drops at its ground pad are allowed.
- Weather at hubs 15, 11, 14, 16: no multirotor launches from
     minute 10 until minute 70.
- A medevac helicopter corridor is active in the corridor
     from (-7.6, 21.0) to (-15.6, 26.9) km, 0.5 km either
     side, below 150 m from minute 16 until minute 76; no
     vehicle may cross it below 150 m.
- Airspace is closed in the circle of radius 6.0 km centered
     at (22.5, 15.0) km from minute 35 until minute 155; no
     flights through it.

Your last orders:
- avoid the circle of radius 6.0 km centered at (22.5, 15.0)
     km until minute 155: applied; flights now detour around
     it
```

```
- prioritize trauma requests until minute 60: applied; those
     requests now come first

Since your last call: 0 delivered (0 on time, 0 late).

Exceptions:
- hub 4: 2 waiting (1 trauma, 1 routine) with no idle vehicle
Region: 16 hubs, 45/80 vehicles idle, 4 requests waiting, 0
     late.

Orders in force:
- flights avoid the corridor from (-7.6, 21.0) to (-15.6,
     26.9) km, 0.5 km either side, below 150 m until minute 76
- flights avoid the circle of radius 6.0 km centered at
     (22.5, 15.0) km until minute 155
- multirotor grounded at hubs 11, 14, 15, 16 until minute 70
- trauma requests first until minute 60
- 11H1 takes drops at its ground pad until minute 70
- 11H2 takes drops at its ground pad until minute 70
- 14H1 takes drops at its ground pad until minute 70
- 14H2 takes drops at its ground pad until minute 70
- 15H1 takes drops at its ground pad until minute 70
- 15H2 takes drops at its ground pad until minute 70
- 16H1 takes drops at its ground pad until minute 70
- 16H2 takes drops at its ground pad until minute 70
- 3H2 takes drops at its ground pad until minute 130

Orders, each a JSON object with one key:
- {"reserve": {"hub": H, "count": N, "site": "CODE",
     "until_min": M}}: keep N vehicles at hub H for that site
     only.
- {"avoid": {"area": {"circle": {"center_km": [x, y],
     "radius_km": r}}, "until_min": M}} or {"avoid": {"area":
     {"corridor": {"from_km": [x, y], "to_km": [x, y],
     "half_width_km": w}}, "until_min": M}}: route every
     flight around the area.
- {"reposition": {"class": "fixed_wing" or "multirotor" or
     null, "count": N, "from_hub": A, "to_hub": B,
     "until_min": M or null}}: fly N idle vehicles from hub A
     to hub B, which becomes their hub; with until_min they
     fly home once idle after that minute.
- {"transfer": {"item": "O-", "units": N, "from_hub": A,
     "to_hub": B}}: fly N units of stock from hub A to hub B.
- {"ground": {"class": "fixed_wing" or "multirotor" or null,
     "hubs": [H, ...], "until_min": M}}: no launches of that
     class (null: any) from those hubs.
- {"prioritize": {"site": "CODE" or null, "urgency": "trauma"
     or "urgent" or "routine" or null, "until_min": M}}:
     serve those requests first.
- {"serve_from": {"site": "CODE", "hubs": [H, ...]}}: let
     those hubs serve the site besides its own.
- {"use_pad": {"site": "CODE", "until_min": M}}: deliver to
     that hospital's ground pad by drop instead of landing on
     its rooftop.

Reply with {"orders": [...]} and nothing else; {"orders": []}
     changes nothing.
```

The reply:

```
{"orders": [{"use_pad": {"site": "16H2", "until_min": 155}},
     {"reposition": {"class": "fixed_wing", "count": 2,
     "from_hub": 5, "to_hub": 4, "until_min": 90}}]}
```

A veto prompt has the same head and, instead of the order format, the replanner's numbered proposals and a one-line instruction. The tail of one such prompt, at minute 41 on a 16-hub day after a stock shortage at five hubs, and its reply:

```
Proposed orders from the dispatcher's planner:
1. transfer 50 units of A+ from hub 6 to hub 11
2. serve 11A from hubs 6
3. serve 11B from hubs 6
4. serve 11C from hubs 6
5. serve 11D from hubs 6
6. serve 11E from hubs 6
7. serve 11F from hubs 6
8. serve 11H1 from hubs 6
9. serve 11H2 from hubs 6
10. transfer 50 units of A+ from hub 3 to hub 7
11. serve 7A from hubs 3
12. serve 7B from hubs 3
13. serve 7C from hubs 3
14. serve 7D from hubs 3
15. serve 7E from hubs 3
16. serve 7F from hubs 3
17. serve 7H1 from hubs 3
18. serve 7H2 from hubs 3
19. transfer 50 units of A+ from hub 6 to hub 10
20. serve 10A from hubs 6
21. serve 10B from hubs 6
22. serve 10C from hubs 6
23. serve 10D from hubs 6
24. serve 10E from hubs 6
25. serve 10F from hubs 6
26. serve 10H1 from hubs 6
27. serve 10H2 from hubs 6
28. transfer 50 units of A+ from hub 8 to hub 12
29. serve 12A from hubs 8
30. serve 12B from hubs 8
31. serve 12C from hubs 8
32. serve 12D from hubs 8
33. serve 12E from hubs 8
34. serve 12F from hubs 8
35. serve 12H1 from hubs 8
36. serve 12H2 from hubs 8
37. transfer 50 units of A+ from hub 14 to hub 15
38. serve 15A from hubs 14
39. serve 15B from hubs 14
40. serve 15C from hubs 14
41. serve 15D from hubs 14
42. serve 15E from hubs 14
43. serve 15F from hubs 14
44. serve 15H1 from hubs 14
45. serve 15H2 from hubs 14

Reply with {"drop": [numbers]}, the proposed orders to drop,
     and nothing else; {"drop": []} keeps them all.
```

```
{"drop": [19]}
```

IV. Per-run counts

Table I gives, for each of the 40 model runs in the paper's model table, the number of model calls, the orders the model gave (or, for the veto, the replanner's proposals it kept), how many were applied and how many refused by the dispatcher, the replies that could not be read, the moves (applied relocations and stock flights), the lending permissions, and the vehicles lost.

TABLE I

PER-RUN COUNTS FOR THE 40 MODEL RUNS. REFUSED ORDERS ARE THOSE THE DISPATCHER COULD NOT APPLY, ALMOST ALL FOR TOO FEW IDLE VEHICLES. UNREADABLE REPLIES WERE NOT JSON IN THE EXPECTED FORM AND CHANGED NOTHING.

Hubs	Role	Model	Seed	Calls	Orders	Applied	Refused	Unreadable	Moves	Lends	Lost
16	Live	Haiku	1	26	44	44	0	0	3	24	0
16	Live	Haiku	2	26	33	33	0	0	1	12	0
16	Live	Haiku	3	27	44	43	1	0	4	18	1
16	Live	Sonnet	1	26	69	62	7	0	29	12	8
16	Live	Sonnet	2	26	60	55	5	1	14	22	5
16	Live	Sonnet	3	27	109	98	11	0	22	59	2
16	Live	Opus	1	26	196	171	25	0	42	108	7
16	Live	Opus	2	26	172	157	15	0	33	84	4
16	Live	Opus	3	27	236	224	12	0	58	124	2
16	Live	GPT-6 Luna	1	26	143	139	4	1	2	60	2
16	Live	GPT-6 Luna	2	26	134	134	0	2	0	76	0
16	Live	GPT-6 Luna	3	27	159	154	5	1	1	101	2
16	Veto	Haiku	1	24	101	101	0	0	9	72	0
16	Veto	Haiku	2	23	115	115	0	0	15	80	0
16	Veto	Haiku	3	23	118	117	1	0	18	78	0
16	Veto	Sonnet	1	23	82	81	1	0	14	48	0
16	Veto	Sonnet	2	24	115	115	0	0	14	80	0
16	Veto	Sonnet	3	23	157	157	0	0	21	116	0
16	Veto	Opus	1	24	151	151	0	0	10	120	0
16	Veto	Opus	2	23	141	141	0	0	16	104	0
16	Veto	Opus	3	23	124	124	0	0	23	80	0
16	Veto	GPT-6 Luna	1	24	122	122	0	0	6	96	0
16	Veto	GPT-6 Luna	2	24	109	109	0	0	1	88	0
16	Veto	GPT-6 Luna	3	24	76	72	4	0	9	43	0
64	Live	Haiku	1	94	176	164	12	0	43	43	17
64	Live	Haiku	2	94	210	198	12	1	24	95	10
64	Live	Sonnet	1	94	382	299	83	2	128	96	16
64	Live	Sonnet	2	94	464	379	85	1	117	188	13
64	Live	Opus	1	94	1043	896	147	0	325	391	38
64	Live	Opus	2	94	950	698	252	0	267	302	34
64	Live	GPT-6 Luna	1	94	1218	1179	39	2	34	597	4
64	Live	GPT-6 Luna	2	94	1674	1631	43	3	5	908	4
64	Veto	Haiku	1	93	596	596	0	0	79	448	0
64	Veto	Haiku	2	93	479	479	0	0	60	352	0
64	Veto	Sonnet	1	91	435	435	0	0	90	265	0
64	Veto	Sonnet	2	94	479	474	5	0	64	328	0
64	Veto	Opus	1	91	472	472	0	0	101	288	0
64	Veto	Opus	2	93	563	563	0	0	64	416	0
64	Veto	GPT-6 Luna	1	93	458	451	7	0	41	328	0
64	Veto	GPT-6 Luna	2	93	625	602	23	0	68	456	0